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Saveetha Institute of Medical And Technical Sciences
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Course Fundamentals of Data Science

Code DSA0404

Slot A

Faculty KRISHNARAJ M

Submitted By Y Sravan Kumar (19247228)

LAB PROGRAMS

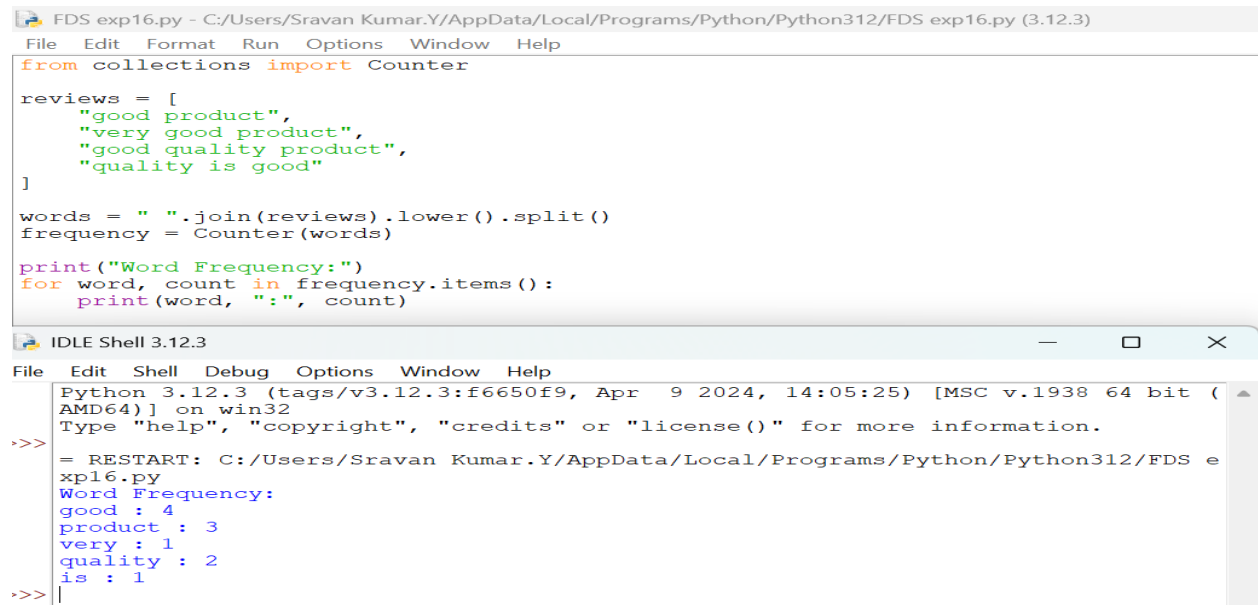


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16. Frequency Distribution of Words in Customer Reviews



The image shows a Python script in an IDE window titled 'FDS exp16.py' and its execution output in an 'IDLE Shell 3.12.3' window. The script uses the 'collections' module's 'Counter' class to count the frequency of words in a list of customer reviews. The reviews are: 'good product', 'very good product', 'good quality product', and 'quality is good'. The output shows the word frequencies: 'good' (4), 'product' (3), 'very' (1), 'quality' (2), and 'is' (1).

```
from collections import Counter

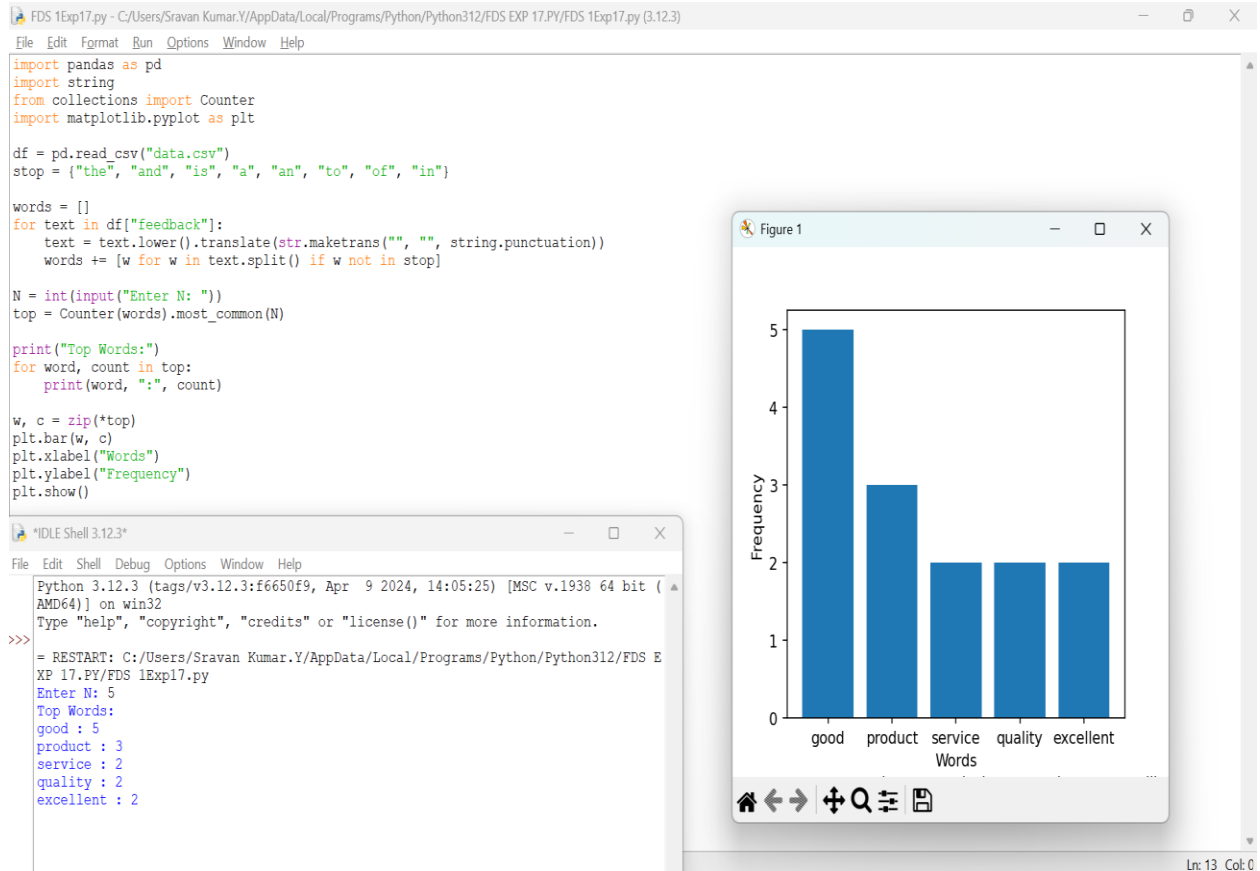
reviews = [
    "good product",
    "very good product",
    "good quality product",
    "quality is good"
]

words = " ".join(reviews).lower().split()
frequency = Counter(words)

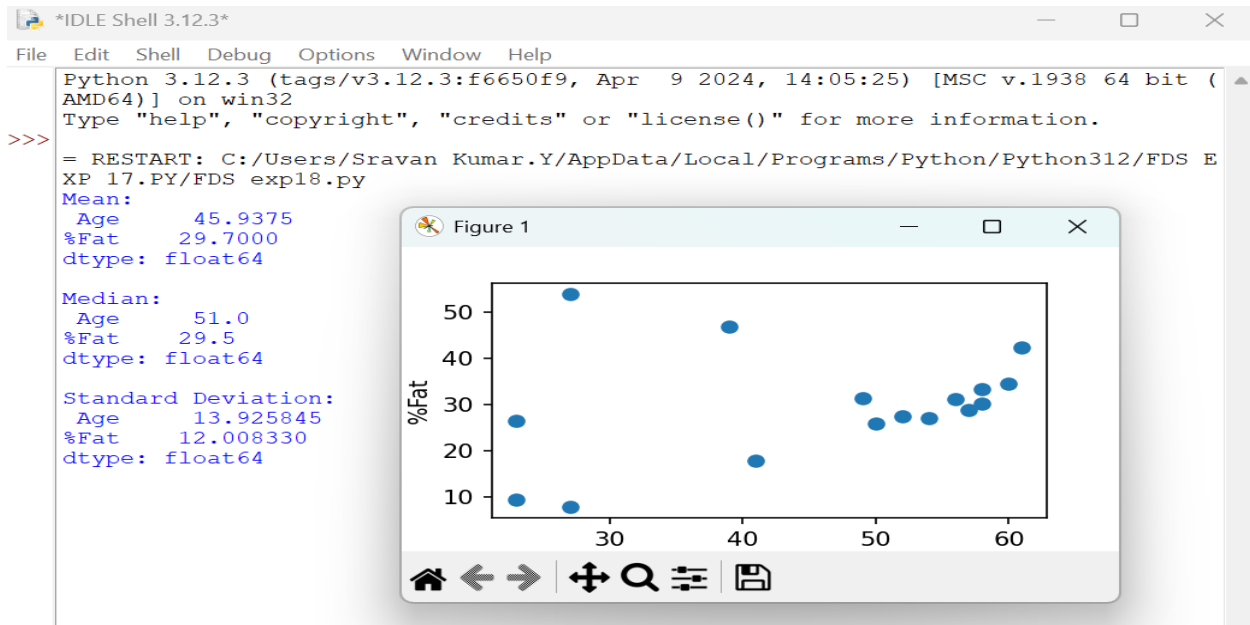
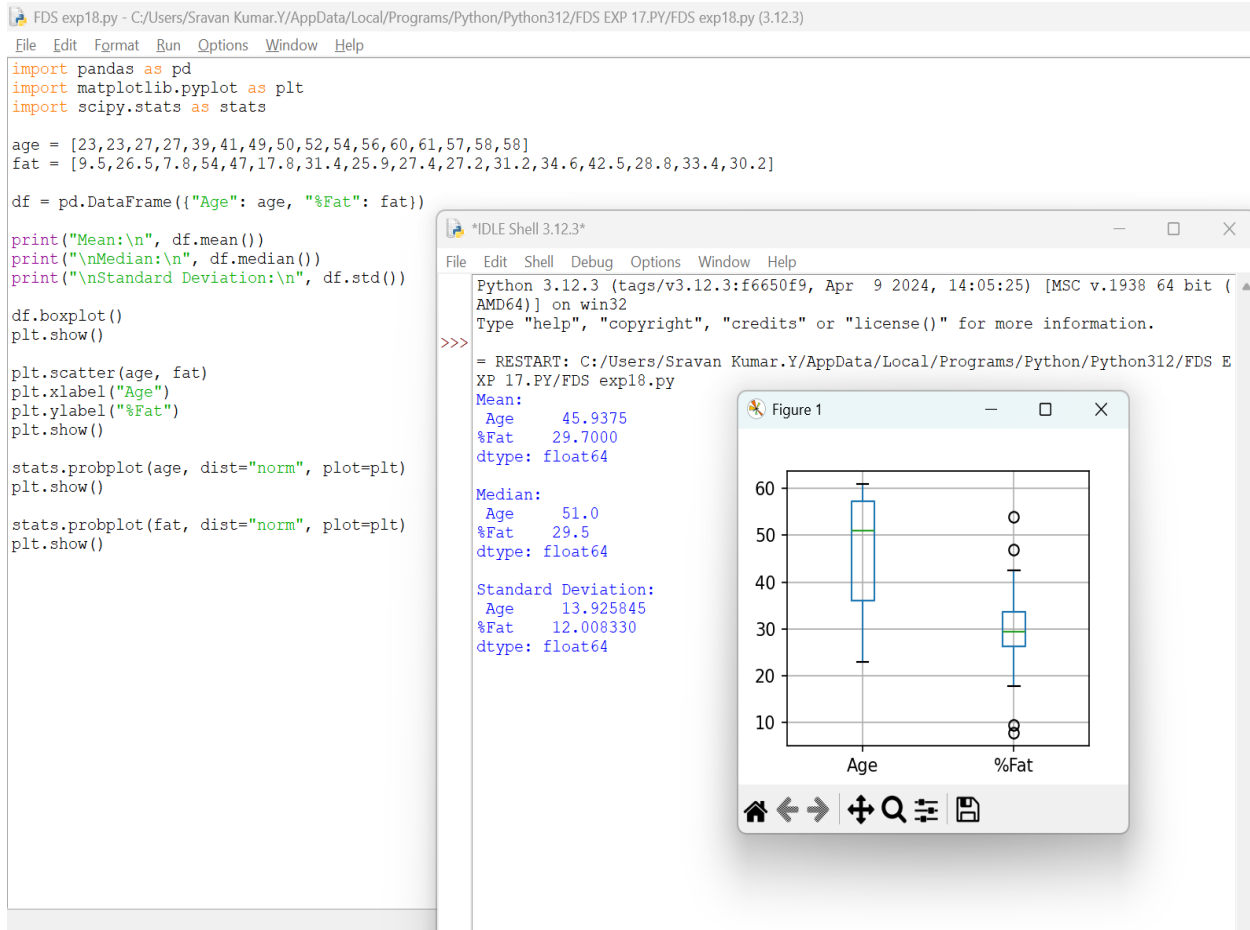
print("Word Frequency:")
for word, count in frequency.items():
    print(word, ":", count)
```

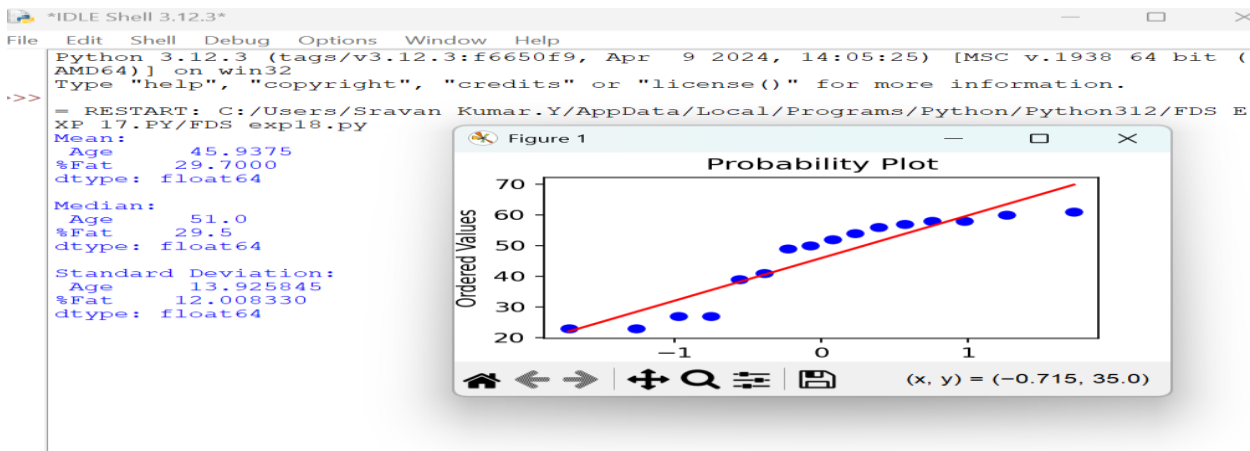
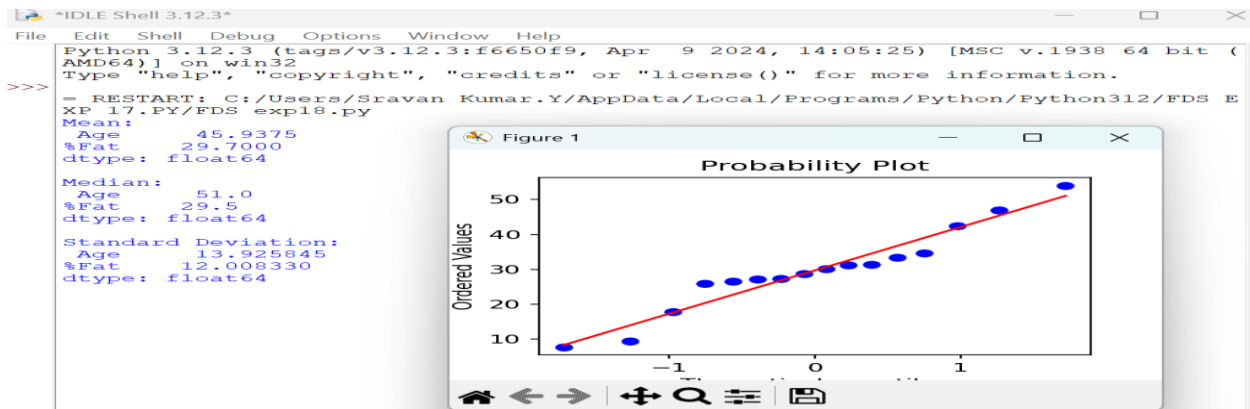
```
Python 3.12.3 (tags/v3.12.3:f6650f9, Apr 9 2024, 14:05:25) [MSC v.1938 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>> = RESTART: C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/FDS exp16.py
Word Frequency:
good : 4
product : 3
very : 1
quality : 2
is : 1
>>>
```

17. Text Preprocessing and Frequency Analysis of Customer Feedback



18. Statistical Analysis and Visualization of Age and Body Fat Data





19. Sales and Profit Analysis Using Pandas

exp 19.py - C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/fds exp 19.py/exp 19.py (3.12.3)

```
File Edit Format Run Options Window Help  
import pandas as pd  
  
# Load the CSV file  
df = pd.read_csv("sales_data.csv")  
  
# Calculate Total Sales  
df["Total Sales"] = df["Quantity Sold"] * df["Unit Price"]  
  
# Calculate profit with 20% profit margin  
df["Profit"] = df["Total Sales"] * 0.20  
  
# Calculate total sales for each product  
product_sales = df.groupby("Product")["Total Sales"].sum()  
  
# Calculate total profit for each product  
product_profit = df.groupby("Product")["Profit"].sum()  
  
# Display total sales  
print("Total Sales for Each Product:")  
print(product_sales)  
  
# Display overall profit  
overall_profit = df["Profit"].sum()  
print("\nOverall Profit:", overall_profit)  
  
# Display top 5 most profitable products  
top5 = product_profit.sort_values(ascending=False).head(5)  
  
print("\nTop 5 Most Profitable Products:")  
print(top5)
```

Python 3.12.3 (tags/v3.12.3:f6650f9, Apr 9 2024, 14:05:25) [MSC v.1938 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.

```
>>> = RESTART: C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/fds e  
xp 19.py/exp 19.py  
Total Sales for Each Product:  
Product  
Keyboard      30000  
Laptop        400000  
Monitor       156000  
Mouse         20000  
Phone         300000  
Tablet        180000  
Name: Total Sales, dtype: int64  
  
Overall Profit: 217200.0  
  
Top 5 Most Profitable Products:  
Product  
Laptop      80000.0  
Phone       60000.0  
Tablet      36000.0  
Monitor     31200.0  
Keyboard    6000.0  
Name: Profit, dtype: float64  
>>>
```

20. Customer Segmentation Based on Total Spending

```
fds exp 20.py - C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/fds exp20.py/fds exp 20.py (3.12.3)
File Edit Format Run Options Window Help

import pandas as pd

# Load the CSV file
df = pd.read_csv("customer_data.csv")

# Function to assign spending segment
def segment_customer(spending):
    if spending >= 50000:
        return "High Spenders"
    elif spending >= 20000:
        return "Medium Spenders"
    else:
        return "Low Spenders"

# Create Segment column
df["Segment"] = df["Total Spending"].apply(segment_customer)

# Display customer segments
print("Customer Segmentation:")
print(df)

# Calculate average age for each segment
average_age = df.groupby("Segment")["Age"].mean()

print("\nAverage Age of Customers in Each Segment:")
print(average_age)
```

```
IDLE Shell 3.12.3
File Edit Shell Debug Options Window Help

Python 3.12.3 (tags/v3.12.3:f6650f9, Apr 9 2024, 14:05:25) [MSC v.1938 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.

>>>
= RESTART: C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/fds exp20.py/fds exp 20.py
Customer Segmentation:
Customer ID Age Gender Total Spending Segment
0 C001 25 Male 80000 High Spenders
1 C002 32 Female 45000 Medium Spenders
2 C003 28 Male 15000 Low Spenders
3 C004 40 Female 70000 High Spenders
4 C005 22 Male 10000 Low Spenders
5 C006 35 Female 55000 High Spenders
6 C007 30 Male 25000 Medium Spenders
7 C008 45 Female 90000 High Spenders
8 C009 27 Male 35000 Medium Spenders
9 C010 38 Female 12000 Low Spenders

Average Age of Customers in Each Segment:
Segment
High Spenders 36.250000
Low Spenders 29.333333
Medium Spenders 29.666667
Name: Age, dtype: float64

>>>
```

Experiment No. 21 – Point Estimation and Confidence Interval

```
exp 21.py - C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/fds exp21.py/exp 21.py (3.12.3)
File Edit Format Run Options Window Help

sample = np.random.choice(data, sample_size, replace=False)

# Point estimate
sample_mean = np.mean(sample)

# Sample standard deviation
sample_std = np.std(sample, ddof=1)

# Standard error
standard_error = sample_std / np.sqrt(sample_size)

# Z-value for common confidence levels
z_values = {
    90: 1.645,
    95: 1.96,
    99: 2.576
}

if confidence not in z_values:
    print("Use 90, 95, or 99 as confidence level.")
    exit()

z = z_values[confidence]

# Margin of error
margin_error = z * standard_error

# Confidence interval
lower = sample_mean - margin_error
upper = sample_mean + margin_error

print("\nSample:", sample)
print("Point Estimate (Sample Mean):", round(sample_mean, 3))
print("Standard Error:", round(standard_error, 3))
print("Margin of Error:", round(margin_error, 3))
print(
    f"({confidence}% Confidence Interval: "
    f"{(lower:.3f), (upper:.3f)})"
)

if margin_error <= precision:
    print("Desired precision is achieved.")
else:
    print("Desired precision is not achieved.")
```

```
IDLE Shell 3.12.3
File Edit Shell Debug Options Window Help

Python 3.12.3 (tags/v3.12.3:f6650f9, Apr 9 2024, 14:05:25) [MSC v.1938 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.

>>>
= RESTART: C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/fds exp21.py/exp 21.py
Enter sample size: 8
Enter confidence level (%): 95
Enter desired precision: 1.0

Sample: [11.8 9.5 12.1 12.3 10.9 10.2 10.4 11.5]
Point Estimate (Sample Mean): 11.088
Standard Error: 0.353
Margin of Error: 0.692
95.0% Confidence Interval: (10.395, 11.780)
Desired precision is achieved.

>>>
```

Ln: 15 Col: 0

22 – Confidence Interval for Customer Ratings

```
exp22.py - C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/fds exp 22.py/exp22.py
File Edit Format Run Options Window Help

import pandas as pd
import numpy as np

# Load customer reviews
df = pd.read_csv("customer_reviews.csv")

# Extract ratings
ratings = df["Rating"]

# Calculate statistics
n = len(ratings)
mean = ratings.mean()
std = ratings.std()

# 95% confidence interval using z = 1.96
z = 1.96

standard_error = std / np.sqrt(n)
margin_error = z * standard_error

lower = mean - margin_error
upper = mean + margin_error

print("Customer Review Analysis")
print("-----")

print("Number of Reviews:", n)
print("Average Rating:", round(mean, 2))
print("Standard Deviation:", round(std, 2))

print("\n95% Confidence Interval:")
print("Lower Limit:", round(lower, 2))
print("Upper Limit:", round(upper, 2))
```

```
IDLE Shell 3.12.3
File Edit Shell Debug Options Wi
Python 3.12.3 (tags/v3.12
AMD64) on win32
Type "help", "copyright",
>>>
= RESTART: C:/Users/Sravan
xp 22.py/exp22.py
Customer Review Analysis
-----
Number of Reviews: 12
Average Rating: 4.29
Standard Deviation: 0.62

95% Confidence Interval:
Lower Limit: 3.94
Upper Limit: 4.64
>>>
```

23 – Hypothesis Testing for Treatment Effectiveness

